Group 27: Rohan Kosalge, Abhishek Dandekar, Bhairavi Senthilkumar, Suhani Vasoya

CY PLAN 101

02 November 2026

Midterm Report: SB 79's Housing Capacity Gap Near Transit

Research Question

Revised Topic

Within SB 79 Transit-Oriented Development (TOD) zones in Alameda and Santa Clara Counties, how much legal housing capacity does SB 79 create? How is that capacity distributed across transit tiers and MTC Equity Priority Communities (EPC)? And what share of it sits on parcels with characteristics associated with near-term redevelopment, indicating where realistic yield is likely to concentrate (soft-site capacity)?

Geography and Time Frame

The geography is parcels inside MTC's SB 79 TOD zones (200 ft, ¼ mile and ½ mile bands around Tier 1 and Tier 2 stops). The midterm results cover Alameda County; Santa Clara County is added for the final. Capacity reflects SB 79 standards and county assessor data as of October 2026. EPC status uses MTC's current EPC tract designations.

What Changed Since the Proposal

Our Berkeley prototype used a bounding box and EPSG:26910. We have replaced it with a county-wide pipeline in EPSG:3310, so every city is processed the same way. This follows MTC's advice to apply one common framework across study areas and then compare the results.

We have also set aside two datasets for now: the GTFS service-frequency index and the HCD Annual Progress Report (Table A2) permit baseline. In their place, the SB 79 tier stands in for transit quality. Tier 1 is heavy rail such as BART, and Tier 2 is light rail, bus rapid transit and qualifying commuter rail. Assessor data stands in for development likelihood, through soft sites and recent recorded transactions. We made this change to get one reliable, reproducible pipeline working first, and we plan to bring the APR data back as a historical baseline if time allows (Section 5).

Following MTC's feedback, we now make an explicit distinction between capacity, the units SB 79 legally allows, and yield, the units likely to be built. Soft-site capacity is our first yield indicator, and it is not a forecast of what will be built. We will present historical activity the same way: as a record of how development happened under prior rules.

Finally, we dropped the idea of ranking stations as first choices for development. Instead, we will compare where SB 79 creates the most new capacity, using four transit-by-EPC typologies, as MTC suggested.

Data

Raw data was retrieved from the **following three sources**: “SB 79 TOD stops & zones” (saved as `tod_zones.geojson`, 665 records), “Alameda County parcels” (saved as `parcels.geojson`, 71,742 records), and “Equity Priority Communities” (saved as `epc_data.geojson`, 353 records). In the `notebooks` folder, we utilize `data_cleaning.ipynb` to pre-process our data and `eda.ipynb` to graph our visualizations and observe relations within our cleaned dataset. The parcel count reflects only parcels intersecting the TOD zones, since we read the over-1 GB county file with a spatial mask. In the notebooks folder, we use 01_data_cleaning.ipynb to pre-process our data and 02_eda.ipynb to graph our visualizations and observe relations within our cleaned dataset.

Methods

To reduce our dataset, we placed several **filters** prior to joining. Here are the most relevant ones:

- removing public land in `parcels.geojson` (21,585 rows removed)
- removing unnecessary “Use Codes” in `parcels.geojson` (these are defined as what sector the parcel is a part of, for example, “industrial” or “agricultural”) (2,645 rows removed)
- keeping only “Santa Clara” and “Alameda” counties in `epc_data.geojson` (183 rows removed)

We proceeded to implement our **spatial joins**:

- **(1) within**: `parcels_data` + `tod_data`. Each parcel’s representative point (a point guaranteed to fall inside the parcel) is matched to the TOD zone it falls within. Where zones overlap, we keep the most permissive density standard.
- **(2) within**: `parcels_data/tod_data + epc_data`. each TOD parcel’s representative point is matched to the EPC tract it falls within.

For **(2)**, we computed a **union** with **(1)** to create a one-hot encoding column labeled `is_epc` (“Does this TOD parcel fit entirely within an EPC area?”). For this geospatial data we used the **CRS 3310**: the “California Albers" projection mapping, as we are working with California-specific data.

Early Results

Using this cleaned, joined dataset, we have observed the following patterns so far:

- As EPC level rises (“high” → “higher” → “highest”), SB 79 capacity shifts from Tier 1 (BART) to Tier 2 (Bus Rapid Transit).
- Noting that a “soft site” is defined as a parcel where the building is worth less than half the land, only a small "slice" of allowed capacity sits on soft sites, and that slice gets smaller as EPC level rises.
- SB 79 allows 688,068 units in total across these parcels (473,262 in Tier 1 zones and 214,806 in Tier 2 zones), and about half of the parcels (50.3%) fall in an EPC. Most capacity sits in the ½-mile bands rather than next to stations.
- Capacity in EPCs leans much more heavily on Tier 2 (bus rapid transit and light rail) than capacity outside EPCs. Tier 2 makes up 11% of allowed units outside EPCs, compared with 47% in “High,” 76% in “Higher” and 64% in “Highest” EPCs.
- Defining a “soft site” as a parcel where the building is worth less than half the land, only 12.9% of allowed capacity sits on soft sites. That share is lower inside EPCs: 15.0% outside EPCs, 13.6% in “High,” 7.1% in “Higher” and 8.5% in “Highest.”

Plans to Finish

All four of us will meet with Kearey Smith at MTC by November 9 to review the midterm findings. By November 13, Rohan and Suhani will add Santa Clara County parcels and run the same pipeline, rewritten as functions in `src/capacity.py`. By November 16, Abhishek and Rohan will estimate net capacity by subtracting existing units, using assessor unit counts or a building-area proxy. By November 18, Bhairavi and Abhishek will test how sensitive the soft-site measure is to its threshold (0.3, 0.5 and 1.0) and compare recent activity between EPC and non-EPC areas. By November 20, Bhairavi and Suhani will build a station-area summary of the capacity gained at each station and group stations into four typologies (Tier 1 or Tier 2, EPC or non-EPC). As a stretch goal, we will add HCD APR Table A2 permits near stations from 2018 to 2025 as a historical baseline by November 25. The whole team will finish the final maps, charts, README and written report by the final due date.

Biggest Risks and Plan to Adapt

1. **Santa Clara parcel data.** The county's parcel file may use different fields for land value, improvement value and use codes. If it lacks those values, we will report legal capacity only and limit the soft-site analysis to Alameda.

2. **No existing-unit counts.** Our parcel file has no unit counts. If we cannot get them, we will use codes to estimate existing units (single-family = 1, duplex = 2, and so on) and label the result as an estimate.

3. **Processing time.** The parcel file is over 1 GB. We already read it with a spatial mask and save cleaned data as Parquet, and we will do the same for Santa Clara.

4. **APR geocoding.** Table A2 records have addresses or APNs, not coordinates. If matching them to parcels takes too long, we will drop the historical baseline and rely on the assessor-based yield indicators.

Areas For Feedback

1. Is the parcel-level EPC assignment (representative point in tract) defensible, or should we also report how much of each TOD buffer overlaps EPCs, as a check?

2. Is the soft-site ratio (improvements < 50% of land value) a reasonable yield indicator, given Proposition 13 assessment lags? Does MTC use a preferred alternative?

3. Do 4 typologies (Tier 1/Tier 2 × EPC/non-EPC) capture the comparisons most useful to MTC, or would a different split work better, such as a typology by distance band?

Works Cited

Alameda County. "Parcels." *Alameda County Open Data Hub*, https://data.acgov.org/datasets/2b026350b5dd40b18ed7a321fdcdba81_0/explore. Accessed 2 Nov. 2026.

Metropolitan Transportation Commission. "Equity Priority Communities - Plan Bay Area 2050 Plus (ACS 2018-2022)." *MTC Open Data*, https://opendata.mtc.ca.gov/datasets/1ae2ccc15b4a4fe9a6d0cd84ae47f2a5_0/explore. Accessed 2 Nov. 2026.

———. "Senate Bill 79 - Transit-Oriented Development Zones." *MTC Open Data*, https://opendata.mtc.ca.gov/datasets/ff3aca5ea60145c0ad677edbbf3f896f_1/explore. Accessed 2 Nov. 2026.